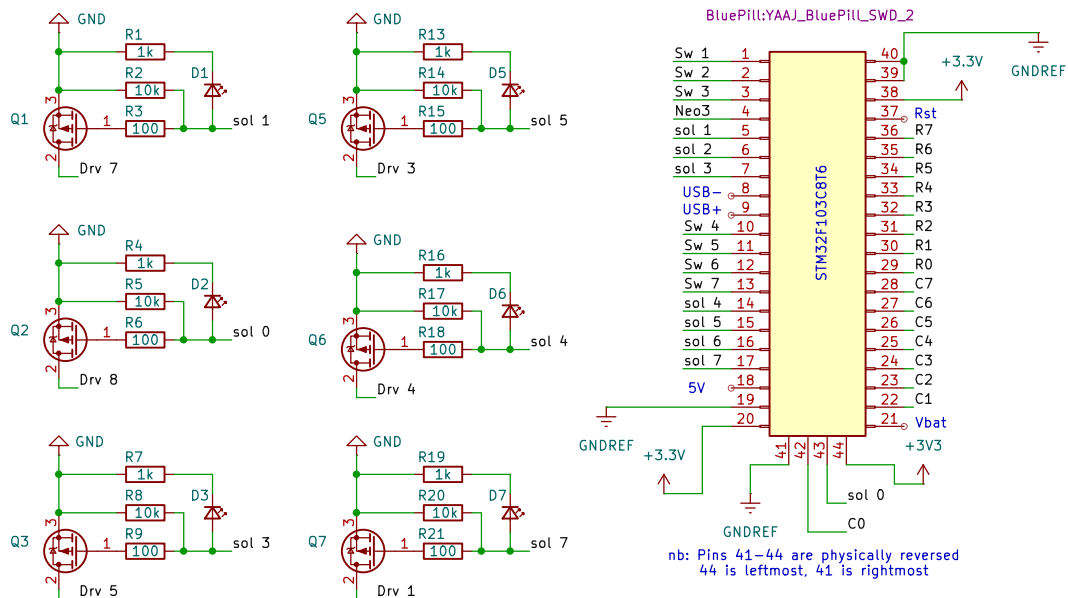
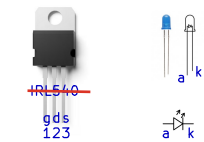
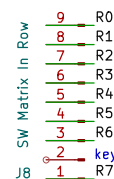
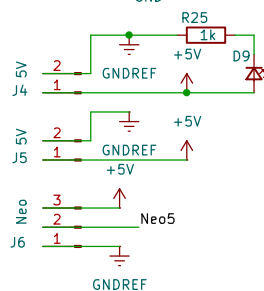
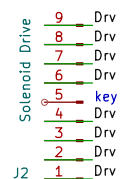
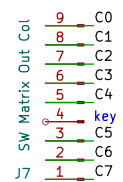
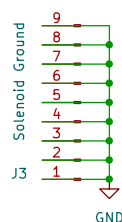
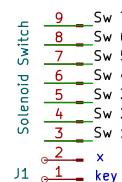
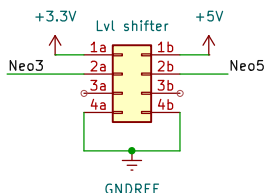
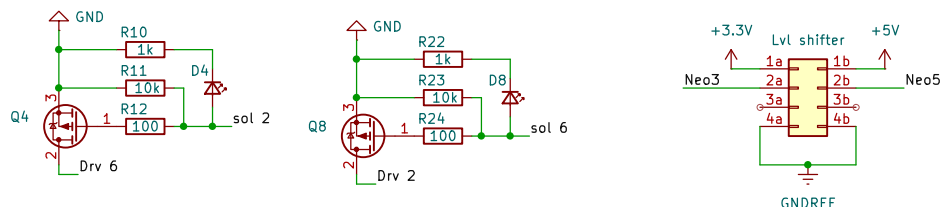




level shifter top view



IRLZ44NPbF



Notes:
GND = solenoid voltage ground
GNDREF = 3V3 and 5V ground
Bluepill ground & 3V3 pins are common

Parts List

Qty	Item
1	STM32F103C8T6
1	3.3V / 5V level shifter *
1	IIC I2C Logic Level Converter
8	IRLZ44NPbF
8	3 mm white led
1	3 mm green led
9	1k resistor
8	10k resistor
8	100r resistor
2	20 pin 2.54 female single header
3	4 pin 2.54 female single header
2	2 pin 3.96mm header
1	3 pin 3.96mm header
1	4 pin 3.96mm header
5	9 pin 3.96mm header

* Level shifter pinout must match the diagram shown.
They come in a variety of configs

Rev 1.1 Mosfets changed to IRLZ44NPbF for lower Vgs switching threshold

Sheet: /
File: SolNeoSWmat.kicad_sch

Title: OPP Solenoid, Neopixel & Switches

Size: A4	Date: 12 June 2026	Rev: 1.1
KiCad E.D.A. 8.0.3		Id: 1/1